Thermal radiation and color

antocciphysics | TEACHER KEY | CPA

Learning target: Use peak wavelength as one piece of evidence about a thermal spectrum.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 13; 3 course days later.*
20° from the normal, on the other side.

02. Explain a worked example (worked)
Hotter curve peaks at shorter wavelength; visible color cannot be inferred from one peak alone without the full spectrum/observer context.

03. Look and explain (model)
6000 K; no, broad spectra.

04. Build the solution (practice)
Hotter peak about half; 970 nm infrared outside visible.

05. Use the idea (reason)
Broad thermal spectrum still has visible redtail; peak need not be in visible band.

06. Choose a method (mixed) *Review: Unit 6, Day 4; 12 course days later.*
Upright. An ideal free-end reflection does not invert the displacement pulse.

07. Exit check (exit)
Halves.

08. STRETCH: Apply it another way (stretch)
Peak wavelength halves; much radiation is infrared.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 13; 3 course days later.*
Image moves 0.3 m toward the mirror from behind it. Separation decreases by 0.6 m.

02. Explain a worked example (worked)
Hotter curve peaks at shorter wavelength; visible color cannot be inferred from one peak alone without the full spectrum/observer context.

03. Represent and interpret (model)
6000 K; no, broad spectra.

04. Solve and justify (practice)
580 nm; 290 nm(UV). Observed color uses whole spectrum and may include emission lines/atmospheric effects.

05. Analyze the reasoning (reason)
Broad thermal spectrum still has visible redtail; peak need not be in visible band.

06. Choose a method (mixed) *Review: Unit 6, Day 4; 12 course days later.*
First $\lambda=6$ m; second $\lambda=8$ m, using $\lambda=v/f$.

07. Exit check (exit)
4000 K.

08. STRETCH: Apply it another way (stretch)
≈967 and 483 nm; broad emission, visual sensitivity and conditions determine appearance, not one spectral wavelength.